

END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: B. Tech Biotechnology

Name of the Paper Enzyme Technology

Time: 3 Hours

Semester: Vth

Paper Code: TBT-502

Maximum Marks: 100

Note:- (i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Question 1		CO
(a)	1. Elaborate the introduction and scope of enzymology. (6) 2. Define the following terms: Coenzyme, Isoenzyme, and Prosthetic group. (4)	CO1
(b)	1. Diagrammatically explain the principle of affinity chromatography in detail. (6) 2. Discuss isolation and purification of enzyme. (4)	
(c)	1. Explain the enzyme specific activity and specific activity. (6) 2. Elaborate in detail on some of the important properties of enzyme. (4)	
Question 2		
(a)	Discuss the problem associated with Michaelis Menten equation and derive the Lineweaver Burk plot equation to overcome this problem. Discuss how these parameters are experimentally determined. (10)	CO2
(b)	Analyze the "lock and key" hypothesis versus the "induced fit" model of enzyme action in detail. (10)	
(c)	1. Given $V_{max}=500 \mu\text{mol/min}$ and $K_m=10 \text{ mM}$, calculate the reaction velocity at a substrate concentration of 15 mM (4) 2. What will be the effect of increase in substrate and enzyme concentration on enzyme velocity? (6)	
Question 3		
(a)	What is activation energy? Explain how enzymes can be used as catalysts to lower the activation energy. (10)	CO3
(b)	What is enzyme regulation? Describe the different mechanisms of enzyme regulation in detail. (10)	
(c)	What do you mean by enzyme inhibition? Explain the different types of reversible inhibition in detail. (10)	
Question 4		
(a)	1. Enzymes can be used in disease therapy. Do you agree? Whether yes/no, give reasons. (5) 2. Explain the role of enzymes in food industry. (5)	CO4
(b)	Describe the concept of artificial enzymes. Provide examples and explain their potential applications. (5)	
(c)	Describe microbial Alpha and β amylase production process? Discuss their industrial applications with examples (10)	
Question 5		
(a)	What do you mean by enzyme engineering? Discuss the various strategies employed in enzyme engineering to improve stability, activity, and specificity? Discuss these approaches with examples from industrial applications. (10)	CO5
(b)	Explain the role of industrially important microorganism isolation, screening & maintenance. (10)	
(c)	Define Immobilized enzymes? Explain the techniques of immobilization in details? (10)	

